

Quantum Physics

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Q1. For a certain metal, when monochromatic light of wavelength λ is incident, the stopping potential for photoelectrons is $3V_0$. When the same metal is illuminated by light of wavelength 2λ , the stopping potential becomes V_0 . The threshold wavelength for photoelectric emission is $\alpha\lambda$. The value of α is ____.

NTA B.Tech · 2 Apr 2026, Shift 1 (Q42)

- (1) 1
- (2) 4
- (3) 2
- (4) 3

✓ **Answer: (2) 4**

Solution:

Einstein's photoelectric equation: $eV = hc/\lambda - hc/\lambda_0$.

For λ : $3V_0e = hc/\lambda - hc/\lambda_0$

For 2λ : $V_0e = hc/(2\lambda) - hc/\lambda_0$

Subtracting: $2V_0e = hc/\lambda - hc/2\lambda = hc/2\lambda$, so $V_0e = hc/4\lambda$.

Substituting back: $hc/4\lambda = hc/2\lambda - hc/\lambda_0$, giving $hc/\lambda_0 = hc/4\lambda$, so $\lambda_0 = 4\lambda$.

Thus $\alpha = 4$.

Source: NTA B.Tech — 2 April 2026, Shift 1 (official NTA release, physics section) · formulaverse.in